Alexandros Phardis

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Education

University of Toronto ***09/24 – 04/28***

B.Sc. in Mathematics & Statistics

- **Putnam Mathematical Competition**: Ranked 1,889th out of 4,329 participants (Top 44%) (Score: 3)
- Coursework in Probability, Statistics, Computer Programming, Data Structures & Analysis, Data Analysis, Machine Learning, Real/Complex Analysis, Advanced Calculus, Linear/Abstract Algebra, ODEs/PDEs, Topology, Number Theory

Experience

Data Science Intern ***05/26 – 08/26***

TD Bank Group | ***Analytics, Insights & Artificial Intelligence (AI2) Team***

- Engineered production-grade causal inference and experimental design frameworks on an enterprise SaaS platform, scaling automated sampling and analytical workflows to 410+ internal users
- Implemented Instrumental Variables (IV) and the Local Average Treatment Effect (LATE) framework to estimate causal effects in non-compliant and endogenous settings where standard A/B testing was invalid
- Automated advanced statistical methodologies within the platform, including Instrumented Difference-in-Differences (iDiD), Cube Sampling, Propensity Score Matching (PSM), and sample-size/power calculations
- Developed an automated AI agent using Shiny for Python to refactor exploratory Jupyter Notebooks into production-ready dashboard applications, reducing deployment workflows from days to hours
- Designed and delivered a technical curriculum (101/201 courses and Learning Lounges) on causal inference and experimental design, training 180+ data analysts and scientists
- Earned Level 1 (Great Work!) and Level 2 (Above & Beyond!) Awards

Teaching Assistant ***08/25***

Canadian Mathematical Society & University of Toronto

- Prestigious invitation-only Canada Math Camp consisting of the top 26 scoring students on the Canadian Open Mathematics Challenge (COMC) in Grade 10 and under
- Students work towards potentially representing Canada in the International Mathematical Olympiad (IMO)
- Supported Olympiad-level problem-solving sessions, oversaw IMO mock tests, and coordinated guest lectures

Projects

No Limit: a Bot Building Tournament (Boston) | *Link to Tournament*, *Link to Code* ***11/26***

- Selected as 1 of 231 teams from a pool of 6,000+ global applicants to compete at Harvard University in a tournament hosted by Sequoia Capital, Citadel Securities, and OpenAI with a $250,000 total prize pool
- Engineered and deployed a live-decision pokerbot with sub-1ms latency constraints during a high-pressure 60-minute coding sprint across >300M simulated hands
- Programmed logic to optimize a 5-card flop auction Hold 'em variant, with pre-computed lookup tables and adaptive opponent-exploitation rules to maximize expected value

Numerical Methods in Scientific Computing | *Link to Project* ***01/26 – 04/26***

- Selective PhD-mentored directed reading program (DRP) from the Math department pairing a group of undergraduates for a proposed project
- Gave weekly presentations on Monte Carlo Simulations, Linear Systems, SVD, ODEs, Integral Equations, Gaussian Quadrature, and more

Trefoil Divisibility: When is a Knot Divisible by the Trefoil? | *Link to Project* ***06/25***

- Joint research project in Knot Theory, under the supervision of a Postdoctoral Fellow and a PhD Candidate
- Presented at the Canadian Undergraduate Mathematics Conference at the University of Waterloo

Leadership

Why Choose Math Campus Day Speaker | *University of Toronto* ***04/26***

- Gave personal insights to high-achieving high schoolers as an invited panel speaker for the Department of Mathematics

COMC (Canadian Open Mathematics Challenge) Marker | *Canadian Mathematical Society* ***11/25***

- Evaluated proofs for the Canadian Mathematical Olympiad (CMO) qualifier, focusing on Part C (Advanced Proofs)

Technical Skills

Languages: Python, R, SQL, LaTeX

Libraries: NumPy, SciPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Statsmodels, Linearmodels, Shiny for Python

Tools: VSCode, PyCharm, Git, GitHub, GitHub Copilot, Claude Code, Codex, Jupyter Notebook